SEGi UNIVERSITY

BCC1313 – ALGORITHMS & COMPLEXITY

Tutorial 2

Answer ALL questions.

1. What is the base case in recursion, and why is it necessary for the recursive function to terminate? How does the absence of a base case affect the program's execution?

2. Explain how the computer manages function calls in recursion using a call stack. How does the call stack grow and shrink during recursive calls, and what might happen if the recursion depth becomes too large?

3. Explain the conceptual difference between recursion and iteration. Can every recursive function be converted into an iterative one?

4. Why might recursion be less efficient than iteration in certain scenarios? Give an example where an iterative approach would be more appropriate due to time or memory constraints.

5. What will be the result of calling the recursive function `myFunction` with the following parameters:
   i) `fun2(5, 3)`
   ii) `fun2(10, 7)`
   iii) `fun2(6, 2)`

   Instructions:
   - Trace the execution of the function for each specified call.
   - Draw a tree diagram representing your solution.

```
int myFunction(int x, int y) {
    if (x <= 0) {
        return 0;
    } else if (y == 0) {
        return 1;
    } else {
        return fun2(x - 1, y - 1) + y;
    }
}
```